

# Ahmed Khaled

[in LinkedIn](#) | [+20 1550366775](#) | [sw.eng.ahmedkhaled@gmail.com](mailto:sw.eng.ahmedkhaled@gmail.com) | [GitHub](#)

## Education

|                                                                 |                            |              |                   |
|-----------------------------------------------------------------|----------------------------|--------------|-------------------|
| Master of Informatics                                           | <u>Nile University</u>     | Giza, Egypt  | 04/2024 - present |
| Bachelor of Computer Science<br>Scientific Computing Department | <u>AinShams University</u> | Cairo, Egypt | 08/2019 - 07/2023 |

- Ranked **FIRST** in Bachelor\*

## Skills

- **Problem Solving:** Data Structures | Algorithms | C++ | python
- **Functional Programming:** sml | racket
- **Databases:** sql | mongoDB | PostgreSQL | NoSQL
- **Tools:** git & Github | bash | Docker
- **Specialized Areas:** Machine Learning | Computer Vision | NLP | Image Processing
- **Languages:** Arabic (Native), English(Proficient)

## Experience

|                                |                        |             |                    |
|--------------------------------|------------------------|-------------|--------------------|
| Research Assistant (full-time) | <u>Nile University</u> | Giza, Egypt | 02/2024 - present! |
|--------------------------------|------------------------|-------------|--------------------|

- Assisted in the NLP Project for Dar Al Iftaa for Automatic Fatwa Generation Model using an Advanced RAG System.
- Have active Interest on the Quantum Computing Field and its Intersection with Machine Learning

|                                |                        |             |                    |
|--------------------------------|------------------------|-------------|--------------------|
| Teaching Assistant (full-time) | <u>Nile University</u> | Giza, Egypt | 02/2024 - present! |
|--------------------------------|------------------------|-------------|--------------------|

- Assist in Algorithms Analysis and Design for Postgraduates
- Assist in teaching CSCI416 Introduction to Big Data, Math212 Discrete Mathematics, CSCI417 Machine Intelligence, CSCI207 Data Structures and Algorithms, CSCI221 Logic Design, Databases for Undergraduates

|                                     |                 |        |                    |
|-------------------------------------|-----------------|--------|--------------------|
| Computer Science Mentor (part-time) | <u>Almentor</u> | online | 12/2023 - present! |
|-------------------------------------|-----------------|--------|--------------------|

- Actively participating in the implementation of the DECI Initiative as presented by the Egyptian Ministry of Communications.

|                               |                          |                                 |                   |
|-------------------------------|--------------------------|---------------------------------|-------------------|
| Coding Instructor (full-time) | <u>Discovery Academy</u> | on-site, Fifth Statement, Egypt | 08/2023 - 01/2024 |
|-------------------------------|--------------------------|---------------------------------|-------------------|

- Design long-term Learning Curricula composed of 10 levels of Programming Experience.

|                           |                  |                      |                   |
|---------------------------|------------------|----------------------|-------------------|
| Software Engineer, Intern | <u>Gensystem</u> | Remote, Saudi Arabia | 07/2023 - 12/2023 |
|---------------------------|------------------|----------------------|-------------------|

- Contributed to the development of "Curious", a Voting System in Saudi Arabia, using Supabase + Flutter Flow Stack, by designing the Database Schema efficiently, and applied Database row-level Security besides an Authentication System.

## Projects

- **Aftina:** Language Model that Leverages RAG System-based that relies on **Trusted** Fatwas Database from Dar Al Iftaa . **[running]**
- **SOCER Graduation Project - Excellent\***  
Retrieving the Exact Time when a soccer action (goal, card, foul) occurs, on which an Automatic Highlighting Generator can be built, using context-aware loss Function (Python, PyTorch, Computer Vision). [\[Code\]](#) [\[Paper\]](#) [\[Documentation\]](#) [\[Demo\]](#). **(07/2023)**
- **Parallel Low Pass Filter:** Using Message Passing Model to implement Distributed Parallel Architecture, applying Convolution Operation on Decomposed Image then Integration. [\[code\]](#) **(05/2023)**
- **FOS:** Developing Principal Modules, Memory Management, CPU Scheduling, Queueing System, and Parallel Execution of Processes(multi-threading) to implement a functional Operating System using C language.[\[code\]](#) **(05/2022)**
- **MIPS:** Simulating CPU by Constructing and Composing essential Logical Components such as Adders, Registers, and MUXs using Verilog. [\[code\]](#) **(12/2021)**
- **Efficient Map Routing:** Graph Search Optimization for specific maps Applications that require frequent Queries beating The naive, classic Dijkstra's shortest path Complexity O(EV).[\[code\]](#) **(9/2021)**

## Mentorship

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- **Problem-Solving Training:** Completed 3 Levels under Coach Supervision. **(01/2019 - 04/2021)**
- **Open Source Society University:** self-taught curriculum in Computer Science, Studied Programming Languages, Computer Science Tools, and Software Design and Architecture. **(01/2024 - 06/2025)**

## Competition Awards

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- Running Kaggle Competition on [White Blood Cell Classification under Severe Class Imbalance](#).
- Represented my College in **The International Collegiate Programming Contest (ECPC)**, 3 consecutive times in **(2020-2021-2022)**.
- **Ranked #1** at Kaggle Machine Learning Completion. **(2022)**

## Certification

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- Huawei Certified HCIA-AI V3. [certified](#). Sep 2022.
- Nvidia Deep Learning Workshop. [certified](#). Sep 2022
- Egfwd Udacity Nanodegree cross-skilling. [certified](#). Aug 2022 - Sep 2022
- ITI - AI Summer Training. [certified](#) Aug 2022 - Sep 2022\

## Youtube Channel

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- Like and Subscribe “) <https://www.youtube.com/@AhmedKhaled-db2tu>